The Dual Duhamel Cocycle and the Origin of Prefix Order Debt

A structural reduction toward the logarithmic upper bound in quartic zero repair

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Abstract

For the quartic zero–repair pair

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}\right), \qquad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64}\right),$$

the matching logarithmic upper bound remains open.

The prefix-refinement part of the problem already admits an exact finite-dimensional dual description: the prefix premium is a dual value gap, an integrated shadow-price work, and its multipliers satisfy a local adjoint recursion. The unresolved question is why increasing causal memory appears numerically to suppress the marginal prefix tax geometrically.

This note identifies a more intrinsic candidate carrier.

The homogeneous part of the normalized adjoint recursion is generated by the convolution operator Q^* and the source shifts S_x^*. These operators commute. Consequently, after t steps, homogeneous propagation depends only on the accumulated symbol sum and is therefore insensitive to the order of the prefix.

All order-sensitive information in the propagated dual state is created by the inhomogeneous normalization and dual-slack sources. An exact discrete Duhamel formula makes this statement explicit.

This leads naturally to a gauge-invariant *dual order-debt cocycle*, obtained by subtracting the conditional expectation with respect to the retained count-plus-suffix memory. The resulting memory innovations form an orthogonal martingale-difference decomposition.

The principal remaining problem is consequently sharpened to a contraction question for these innovations: does one additional unit of causal memory contract the order-sensitive Duhamel tail by a uniform factor strictly smaller than one?

A positive answer, together with a quantitative shadow-work coercivity estimate, would yield the required geometric marginal-tax bound and hence the logarithmic upper bound.

1 The quartic zero–repair setting

Let

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}\right), \qquad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64}\right)$$

on

$$\{0, 1, 2, 3, 4\}.$$

Their generating polynomials satisfy

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.$$

A zero–repair tree is a family of laws

$$\{B_h\}$$

indexed by source histories

$$h = (x_1, \dots, x_t)$$

and satisfying the local causal identity

$$\boxed{Q * B_h = \sum_{x=0}^{4} P(x)\, \delta_x * B_{hx}.} \tag{ZR}$$

The finite-horizon bill is denoted by

$$\mathcal{B}_n.$$

The rigorous lower bound

$$\mathcal{B}_n = \Omega(\log n)$$

is already available. The matching estimate

$$\mathcal{B}_n = O(\log n)$$

is the remaining global objective.

The prefix-refinement premium arising in the dyadic decomposition is already known, on finite truncations, to admit both an exact dual value-gap representation and the homotopy formula

$$\Pi_{n,M} = \int_0^1 \langle \lambda_s, r_0 \rangle \, ds. \tag{1}$$

Thus the prefix problem is genuinely a problem about the propagation of dual shadow prices.

2 The exact local adjoint recursion

Let

$$p(h) := \prod_{i=1}^{|h|} P(x_i), \qquad p(\varnothing) = 1.$$

Let λ_h denote the multiplier of the local zero–repair equality and let a_h denote the normalization multiplier.

Normalize by history probability:

$$u_h := \frac{\lambda_h}{p(h)}, \qquad \beta_h := -\frac{a_h}{p(h)}.$$

Define

$$(Q^* u)(j) = \sum_y Q(y) u(j + y),$$

and

$$(S_x^* u)(j) = u(j + x).$$

At a non-root node

$$h = gx,$$

finite-dimensional dual feasibility takes the local form

$$\boxed{-\beta_h + Q^* u_h - S_x^* u_g \geq 0.} \tag{2}$$

This relation is exact on every finite truncation.
Define its nonnegative slack by

$$\sigma_h := -\beta_h \mathbf{1} + Q^* u_h - S_x^* u_g \geq 0.$$

Equivalently,

$$\boxed{Q^* u_h = S_x^* u_g + \eta_h,} \tag{3}$$

where

$$\boxed{\eta_h := \beta_h \mathbf{1} + \sigma_h.} \tag{4}$$

We refer to η_h as the *local dual source injection.*
It contains precisely the terms absent from the homogeneous adjoint recursion.

3 The homogeneous recursion is order blind

Set

$$R := Q^*, \qquad T_x := S_x^*.$$

Since

$$R = \sum_y Q(y) T_y,$$

and shifts commute,

$$T_x T_y = T_{x+y},$$

we have

$$\boxed{RT_x = T_x R \qquad \text{for every } x.} \tag{5}$$

This elementary identity is the key algebraic observation.

Proposition 1 (Order blindness of homogeneous propagation)**.** *Assume that along a history*

$$h_t = (x_1, \ldots, x_t)$$

the source injections vanish:

$$\eta_{h_k} = 0, \qquad k = 1, \ldots, t.$$

Then

$$\boxed{R^t u_{h_t} = T_{x_1+\cdots+x_t} u_\varnothing.} \tag{6}$$

In particular, the propagated homogeneous dual state depends only on the accumulated symbol sum

$$x_1 + \cdots + x_t$$

and not on the order of the symbols.

Proof. From (**??**) with $\eta_h = 0$,

$$R u_{gx} = T_x u_g.$$

Iterating and using (**??**),

$$\begin{aligned} R^t u_{h_t} &= T_{x_t} R^{t-1} u_{h_{t-1}} \\ &= T_{x_t} T_{x_{t-1}} R^{t-2} u_{h_{t-2}} \\ &= \cdots \\ &= T_{x_t} \cdots T_{x_1} u_\varnothing \\ &= T_{x_1+\cdots+x_t} u_\varnothing. \end{aligned}$$

□

Remark 2. *The conclusion concerns the propagated coordinate*

$$R^t u_h,$$

not necessarily the raw multiplier u_h.

This distinction is essential because Q^* need not be injective. In particular, the quartic convolution operator possesses nontrivial null directions. Thus one must not infer from (**??**) *that the raw dual multiplier is itself determined by the accumulated count.*

4 An exact discrete Duhamel formula

The inhomogeneous recursion (**??**) can be solved exactly.

For

$$h_t = (x_1, \dots, x_t),$$

write

$$h_k = (x_1, \dots, x_k)$$

and define the suffix sum

$$S_{k+1:t} := x_{k+1} + \cdots + x_t,$$

with

$$S_{t+1:t} := 0.$$

Theorem 3 (Dual Duhamel cocycle)**.** *For every history h_t,*

$$\boxed{R^t u_{h_t} = T_{x_1+\cdots+x_t} u_\varnothing + \sum_{k=1}^{t} T_{S_{k+1:t}} R^{k-1} \eta_{h_k}.} \tag{7}$$

Proof. The case $t = 1$ is exactly (**??**).

Assume the formula at depth $t-1$. Then

$$R u_{h_t} = T_{x_t} u_{h_{t-1}} + \eta_{h_t}.$$

Applying R^{t-1},

$$R^t u_{h_t} = R^{t-1} T_{x_t} u_{h_{t-1}} + R^{t-1} \eta_{h_t}.$$

By commutativity,

$$R^{t-1} T_{x_t} = T_{x_t} R^{t-1}.$$

Insert the induction hypothesis and combine the shifts. This yields (**??**).

□

Equation (**??**) is the central structural identity of this note.

It separates the propagated dual state into

$$\boxed{\text{order-blind homogeneous transport} + \text{inhomogeneous Duhamel source cocycle.}}$$

The first term cannot distinguish two histories with the same symbol counts.

Therefore:

Corollary 4 (Origin of propagated order sensitivity)**.** *All order-sensitive dependence of*

$$R^{|h|} u_h$$

is generated by the source injections

$$\eta_{h_k} = \beta_{h_k} \mathbf{1} + \sigma_{h_k}$$

and their subsequent propagation.

This is an exact algebraic statement, not a numerical hypothesis.

5 Gauge invariance of the propagated order debt

The dual multiplier possesses the exact additive gauge

$$u_h \sim u_h + c\mathbf{1}.$$

Because Q is a probability law,

$$R\mathbf{1} = \mathbf{1},$$

and therefore

$$R^t(u_h + c\mathbf{1}) = R^t u_h + c\mathbf{1}.$$

Any projection which fixes constants consequently removes the gauge component.

This motivates working directly with the propagated coordinate

$$\boxed{U_h := R^{|h|} u_h.} \tag{8}$$

Unlike a raw norm of u_h, the order-sensitive component of U_h can be made intrinsically gauge invariant.

6 Causal memory and the exact order-debt cocycle

Let

$$H_t = (X_1, \dots, X_t), \qquad X_i \overset{\text{iid}}{\sim} P.$$

For $0 \le K \le t$, define the retained-memory signature

$$\mathfrak{s}_{t,K}(h) = \big(\mathrm{count}(h), \mathrm{suffix}_K(h)\big).$$

Let

$$\mathcal{F}_{t,K} := \sigma(\mathfrak{s}_{t,K}(H_t))$$

be the corresponding information sigma-field.

These fields are nested:

$$\mathcal{F}_{t,0} \subseteq \mathcal{F}_{t,1} \subseteq \cdots \subseteq \mathcal{F}_{t,t}.$$

Let

$$\Pi_{t,K} := \mathbb{E}[\cdot \mid \mathcal{F}_{t,K}]$$

denote conditional expectation in the history coordinate.

Definition 5 (Dual prefix order debt)**.** *Define*

$$\boxed{\mathcal{D}_{t,K} := (I - \Pi_{t,K}) U_{H_t} = (I - \Pi_{t,K}) R^t u_{H_t}.} \tag{9}$$

The homogeneous term in (**??**) is a function only of

$$X_1 + \cdots + X_t,$$

hence is already measurable with respect to the count information in $\mathcal{F}_{t,0}$.

Therefore it disappears under

$$I - \Pi_{t,K}.$$

Theorem 6 (Exact prefix-order cocycle)**.** *For every t and $K \le t$,*

$$\boxed{\mathcal{D}_{t,K} = (I - \Pi_{t,K}) \sum_{k=1}^{t} T_{X_{k+1}+\cdots+X_t} R^{k-1} \eta_{H_k}.} \tag{10}$$

In particular, the order debt contains no homogeneous contribution.

Proof. Apply $I - \Pi_{t,K}$ to (**??**). The homogeneous term

$$T_{X_1+\cdots+X_t} u_\varnothing$$

is $\mathcal{F}_{t,0}$-measurable and therefore also $\mathcal{F}_{t,K}$-measurable. Hence its projection residual is zero. □

Equation (**??**) gives a precise version of the informal statement

$$\boxed{\text{prefix order debt is generated by local dual sources, not by homogeneous transport.}}$$

More precisely, homogeneous transport can propagate an existing source, but it does not create new order sensitivity.

7 Memory innovations

The nested memory filtration gives a canonical decomposition of the order debt.

Define the K-th memory innovation by

$$\boxed{\delta_{t,K} U := (\Pi_{t,K+1} - \Pi_{t,K}) U_{H_t}.} \tag{11}$$

On a finite truncation, equip the buffer coordinate with any fixed Hilbert norm and use the corresponding

$$L^2(P^{\otimes t})$$

norm in the history variable.

Since conditional expectations onto nested sigma-fields are orthogonal projections, the innovations are mutually orthogonal.

Proposition 7 (Orthogonal memory decomposition)**.** *For $K < t$,*

$$\boxed{\mathcal{D}_{t,K} = \sum_{\ell=K}^{t-1} \delta_{t,\ell} U.} \tag{12}$$

Moreover,

$$\boxed{\|\mathcal{D}_{t,K}\|_2^2 = \sum_{\ell=K}^{t-1} \|\delta_{t,\ell} U\|_2^2.} \tag{13}$$

Proof. Since full suffix memory determines the history,

$$\Pi_{t,t} U = U.$$

Hence

$$U - \Pi_{t,K} U = \sum_{\ell=K}^{t-1} (\Pi_{t,\ell+1} - \Pi_{t,\ell}) U.$$

Orthogonality is the standard martingale-difference orthogonality for nested conditional expectations. □

This identifies a canonical scale-by-scale quantity which was absent from the previous prefix formulation.

8 Candidate energy for forgotten order

For a depth-r dual witness define, for example,

$$\boxed{\mathcal{E}_{r,K} := \sum_{t=K+1}^{r} w_{r,t} \, \|\delta_{t,K} U\|_2^2,} \tag{14}$$

where

$$w_{r,t} \geq 0$$

are deterministic weights to be chosen according to the exact telescoping or shadow-work formula.

An alternative tail energy is

$$\boxed{\widetilde{\mathcal{E}}_{r,K} := \sum_{t=K}^{r} w_{r,t} \, \|\mathcal{D}_{t,K}\|_2^2.} \tag{15}$$

By (**??**), these two descriptions are related by an exact orthogonal tail decomposition.

The crucial point is that these quantities are:

(i) gauge invariant;

(ii) genuinely order sensitive;

(iii) canonically associated with suffix memory;

(iv) constructed from the exact adjoint recursion;

(v) zero on purely count-dependent propagated dual states.

Thus they satisfy the structural requirements previously imposed on a candidate carrier of the marginal prefix tax.

9 What should contract

A tempting but incorrect target would be a uniform spectral gap for Q^* alone.

One should not expect an estimate of the form

$$\|R^K\|_{\mathrm{quot}} \leq C\rho^K, \qquad \rho < 1,$$

uniformly over growing support truncations.

The convolution operator possesses slowly varying near-invariant directions, and its isolated spectral geometry is not the geometry of the complete tree-coupled prefix problem.

The candidate contraction must involve both

$$\boxed{\text{Duhamel source propagation}}$$

and

$$\boxed{\text{successive memory projections.}}$$

This suggests the following sharpened problem.

Problem 8 (Memory-innovation contraction)**.** *Find a cutoff-stable Hilbert norm, constants*

$$C < \infty, \qquad 0 < \rho < 1,$$

and a canonical admissible selection or set-valued formulation of the dual witnesses such that

$$\boxed{\mathcal{E}_{r,K+1} \le \rho^2 \mathcal{E}_{r,K} + R_{r,K},} \tag{16}$$

where the remainder $R_{r,K}$ is either zero or summable/absorbable into an $O(K)$ startup term, uniformly in the horizon r.

In the ideal homogeneous form,

$$\boxed{\mathcal{E}_{r,K+1} \le \rho^2 \mathcal{E}_{r,K}.} \tag{17}$$

The previously observed numerical contraction ratio near 0.63, if genuine, should therefore be sought as a contraction of the order-sensitive Duhamel cocycle rather than as a bare eigenvalue of Q^*.

10 The remaining no-hiding problem

Identifying the cocycle does not yet prove that it carries the prefix premium.

The exact homotopy identity is

$$\Pi_{n,M} = \int_0^1 \langle \lambda_s, r_0 \rangle \, ds.$$

A quantitative comparison is still required between this shadow work and the memory-innovation energy.

A schematic sufficient estimate would be

$$\boxed{\Pi_{n,M}^q \le C_0 + C_1 \mathcal{E}_{n,K}} \tag{18}$$

for some $q > 1$, or a direct marginal estimate

$$\boxed{D_{r+1}^{(K)} - D_r^{(K)} \le C\, \mathcal{E}_{r,K}^{1/2}.} \tag{19}$$

This is the surviving quantitative no-hiding problem.

The conceptual advantage of (**??**) is that the object to which such an estimate should be applied is no longer anonymous.

It is the order-sensitive component of an explicit Duhamel source cocycle.

11 Conditional logarithmic closure

The preceding reduction can be summarized in a clean conditional statement.

Theorem 9 (Conditional closure through the Duhamel order debt)**.** *Assume there exist constants*

$$A, B, C < \infty, \qquad 0 < \rho < 1,$$

such that for all relevant r, K:

(i) the retained-memory startup cost satisfies

$$C_{\mathrm{base}}(K) \le A + BK;$$

(ii) the order-sensitive Duhamel energy contracts geometrically,

$$\mathcal{E}_{r,K} \le C\rho^{2K};$$

(iii) the one-step marginal horizon tax obeys the no-hiding estimate

$$D_{r+1}^{(K)} - D_r^{(K)} \le C\,\mathcal{E}_{r,K}^{1/2}.$$

Then

$$\boxed{D_r^{(K)} \le A + BK + Cr\rho^K.} \tag{20}$$

Consequently, choosing

$$K = \left\lceil \frac{\log r}{\log(1/\rho)} \right\rceil + O(1)$$

gives

$$D_r^{\mathrm{full}} = O(\log r).$$

Via the established cut/glue bridge from the fixed terminal target, this yields

$$\boxed{\mathcal{B}_n = O(\log n).}$$

Together with the existing lower bound,

$$\boxed{\mathcal{B}_n = \Theta(\log n).}$$

Proof. From (ii)–(iii),

$$D_{r+1}^{(K)} - D_r^{(K)} \le C\rho^K.$$

Summing over the horizon gives

$$D_r^{(K)} \le A + BK + Cr\rho^K.$$

Since the unrestricted problem is no more expensive than the K-memory restriction,

$$D_r^{\mathrm{full}} \le D_r^{(K)}.$$

Choosing $K \asymp \log r$ makes

$$r\rho^K = O(1)$$

while

$$BK = O(\log r).$$

The cut/glue implication then transfers the logarithmic estimate to the original bill.

□

12 Logical status

The purpose of this note is not to promote the logarithmic upper bound prematurely.

The current status is:

Statement	Status
Finite-truncation dual value-gap identity for the prefix premium	**PROVED**
Integrated shadow-work formula	**PROVED**
Local normalized adjoint recursion $$-\beta_h + Q^* u_h - S_x^* u_g \geq 0$$	**PROVED**
Commutation $$Q^* S_x^* = S_x^* Q^*$$	**PROVED**
Exact dual Duhamel formula (**??**)	**PROVED**
Homogeneous propagated state is order blind	**PROVED**
Exact order-debt identity (**??**)	**PROVED**
Orthogonal suffix-memory innovation decomposition	**PROVED**
Uniform contraction $$\mathcal{E}_{r,K+1} \leq \rho^2 \mathcal{E}_{r,K}, \quad \rho < 1$$	**OPEN**
Uniform no-hiding estimate linking $\mathcal{E}_{r,K}$ to the prefix premium	**OPEN**
Geometric marginal-tax bound $$D_{r+1}^{(K)} - D_r^{(K)} \leq C\rho^K$$	**TARGET**
Matching upper bound $$\mathcal{B}_n = O(\log n)$$	**OPEN**
Two-sided law $$\mathcal{B}_n = \Theta(\log n)$$	**OPEN**

13 The next falsification experiment

The present reduction suggests a much more specific numerical test than another scalar fit of $D_r^{(K)}$.

For every solved primal–dual cell (r, K), export the normalized dual field u_h and compute

$$U_h = (Q^*)^{|h|} u_h.$$

Then form the exact history projections

$$\Pi_{t,K} U = \mathbb{E}[U \mid \mathrm{count}(H_t), \mathrm{suffix}_K(H_t)].$$

Measure the innovation energies

$$\|\delta_{t,K}U\|_2, \qquad \delta_{t,K} = \Pi_{t,K+1} - \Pi_{t,K},$$

and aggregate them according to (**??**).

The pre-registered test is:

$$\boxed{\frac{\mathcal{E}_{r,K+1}}{\mathcal{E}_{r,K}} \approx \rho^2} \tag{21}$$

with the same ratio across several horizons r.

The hypothesis must be rejected if

(F1) the ratio depends materially on r;

(F2) no cutoff-stable norm produces contraction;

(F3) the innovation energy contracts but fails to track the marginal prefix tax;

(F4) the apparent contraction disappears when the entire optimal-dual face is audited rather than one selected optimizer;

(F5) the required projection or normalization is not compatible with the exact dual gauge.

A stable decimal ratio alone is not a theorem.

The desired endpoint is an algebraic inequality or a finite rational operator

$$T$$

acting on the innovation state, with

$$\rho(T) < 1.$$

14 Conclusion

The prefix problem can now be reformulated more sharply.

The homogeneous normalized dual recursion

$$Q^* u_{gx} = S_x^* u_g$$

does not create order information: after propagation by the appropriate power of Q^*, it depends only on the accumulated source symbols.

The order-sensitive component is created entirely by the local inhomogeneous sources

$$\eta_h = \beta_h \mathbf{1} + \sigma_h$$

and is propagated through an exact Duhamel cocycle.

After quotienting by retained causal memory, this produces the canonical order-debt field

$$\mathcal{D}_{t,K} = (I - \Pi_{t,K})(Q^*)^t u_{H_t}.$$

Thus the previous question

"What carries the marginal prefix tax?"

has been reduced to the more concrete question

Does one additional unit of causal memory uniformly contract the order-sensitive Duhamel source cocycle, and does the resulting innovation energy quantitatively control the shadow work required to restore the forgotten prefix constraints?

This is now the load-bearing prefix problem.